Mahathi Sharma

Personal Data

Date of birth	30 May 1997 **Nationality:** India
Location	Bengaluru, India
Email	mahathisharmaa@gmail.com

Education

Jul 2015 – Aug 2022
Navi Mumbai, India

Master of Technology (Integrated) – Biotechnology | *D.Y. Patil Deemed to be University, School of Biotechnology and Bioinformatics*

Grade (final year): 81.66%

Courses: Molecular Markers and RNA Technology, Nanotechnology, Stem cell and tissue engineering, Genomics and Proteomics

M.Tech Thesis: NCoR1 mediated regulation of mitochondrial function by *Mycobacterium tuberculosis* in macrophages.

Professional and Research Experience

Jan 2026 – Present
Bengaluru, India

Research Assistant | *Indian Institute of Science (IISc)*

- Standardizing protocol for Dengue virus rescue from pFK-DV plasmid

Dec 2023 – Nov 2025
Bengaluru, India

Research Associate | *Mynvax Private Limited*

- Maintenance of cell lines such as MDCK and HEp-2 for virus propagation
- Influenza Virus stock perpetuation through regular propagation and calculation of titers using Hemagglutination Inhibition Assay (HAI) and TCID50 assay.
- Serological assays: Hemagglutination Inhibition (HI) Assay, Enzyme-linked Immunosorbent Assay (ELISA).
- Immunization of golden hamster and BALB/c mice for efficacy studies of RSV and Influenza Vaccines at Indian Institute of Science's Central Animal Facility.
- SOP optimization of qPCR and RT-qPCR to estimate viral titers of Baculovirus and Rhabdovirus respectively.

Jul 2023 – Nov 2023
New Delhi, India

Project Associate | *CSIR – Institute of Genomics and Integrative Biology (IGIB)*

Project: Phenome India – CSIR Health cohort knowledgebase.
Position of Responsibility: In-Charge of Sequencing facility at IGIB that houses the NovaSeq 6000, MiSeq, Covaris s220.

- Standardized pan CSIR protocol for human faecal sample collection and processing to look into Gut Microbiome diversity and health.
- Isolated genomic DNA from Human faecal samples to generate sequencing quality DNA.
- Whole Genome Sequencing (WGS) library preparation for the NovaSeq 6000 Illumina Platform.

Jan 2023 – Jun 2023
Chandigarh, India

Project Associate under the guidance of Dr. Pawan Gupta, Chief Scientist | *CSIR – Institute of Microbial Technology (IMTech)*

Project: Exploring host nuclear receptors as targets for host directed adjunct therapy to anti-tuberculosis drugs.

- Performed Western Blotting for checking expression of nuclear receptors (NRs).
- Employed EMSA assay to check protein-DNA binding interaction in NRs.
- Maintained HEK293T, SH-SY5Y and HepG2 cell lines for Transfection of cells.
- Identified the phenotype of a cell in conjunction with its co-expression of other markers in HEK293T and HepG2 cell line with FITC labelled peptides using Confocal Microscopy.
- Worked in a Biosafety level 3 facility for nebulization of mice with *Mycobacterium tuberculosis* culture.
- Vivisected *Mycobacterium tuberculosis* infected mice for the purpose of isolating lungs for IHC and streaking of infected lungs for calculating CFU.

Aug 2022 – Dec 2022
Chandigarh, India

Research Intern under the guidance of Dr. Pawan Gupta, Chief Scientist | *CSIR – Institute of Microbial Technology (IMTech)*

- Observed and learnt techniques such as Western Blotting, EMSA assay, Confocal microscopy.
- Routine maintenance of cell lines such as HEK293T, HepG2 and SH-SY5Y.
- Worked in a BSL-3 facility for handling of *Mycobacterium tuberculosis*.
- Handled mice breeding and genotyping.

Jan 2022 – Jul 2022
Bhubaneswar, India

Project Trainee under the guidance of Dr. Sunil Raghav, Scientist F | *BRIC – Institute of Life Sciences (ILS)*

Project: NCoR1 mediated regulation of mitochondrial function by *Mycobacterium tuberculosis* in macrophages.

- Generated Macrophage specific NCoR1 knockout mice by implementing breeding strategies through crossing of Dendritic Cell and Macrophage specific Double Knockout mice and NCoR1 Floxed mice.

- Screened mice genotype through fast DNA extraction protocol, Agarose Gel Electrophoresis and visualized bands using a gel documentation system.
- Anatomized mice to isolate and examine lungs, heart, spleen, liver, intestines, colon, axillary and mesenteric lymph nodes, femur and tibia.
- Isolated and collected primary cells from bone marrow of femur and tibia to differentiate into macrophages.
- Performed Flow Cytometry analysis to determine phenotype of bone marrow derived cells on differentiation.
- Analysed mRNA level expression of genes using RT-qPCR.
- Identified expression levels of NCoR1 corepressor through Western Blotting.
- Performed Cell Mito Stress test in Seahorse Extracellular Flux Analyzer (XFp) to determine key respiratory parameters in mouse bone marrow-derived macrophages.
- Learnt good lab practices and maintenance of primary cell culture of BMDM cells and cell lines of HEK293T and LLC1.
- Worked in a Biosafety level 2 and 3 facility for culturing and infection of *Mycobacterium tuberculosis* to BMDM cells.

Aug 2021 – Sep 2021
Bengaluru, India

Intern | *Genotypic Technology*

- Received training in extraction of quality genomic DNA for downstream sequencing.
- Observed Next Generation Sequencing (NGS) library preparation for Illumina sequencing and Nanopore sequencing.
- Trained in basic Linux for analysis of NGS data and quality check.

Dec 2019 – Present
India

Academic Tutor | *Self-Employed*

- Facilitating personalized and effective coaching in Mathematics and Science for secondary and senior secondary students.

Publication

Jun 2024

Blood Advances; DOI ID: https://doi.org/10.1182/bloodadvances.2023012228
"Nuclear Receptor Rev-erbA Role in Fine-Tuning Erythropoietin Gene Expression"
Impact Factor: 7.4

Skill Summary

Programming Languages	Python, C++, Perl (BioPerl), R Programming, PyTorch, TensorFlow
Software	GraphPad Prism, Swiss-PdbViewer, SPSS Statistics, AutoDock, ChemSketch, HyperChem, MS Office
Wet Lab	PCR, RT-qPCR, Western Blotting, EMSA, ELISA, Animal Handling, Virus and Cell Culture, BSL-2/3

Languages

Certificates	**Goethe-Zertifikat, Level A2** – exam 26/06/2026, certified 03/07/2026 **Goethe-Zertifikat, Level A1** – exam 27/04/2026, certified 30/04/2026 **IELTS**, CEFR Level C1 – exam 06/12/2025, certified 07/12/2025
Proficiency	English: C1 German: A2 Hindi, Tamil, Kannada, Marathi: Native French: A1

Volunteer Experience

Jul 2020 – Jun 2021
Mumbai, India

Director of Community Service and Joint Director of Partners-in-Service | *Rotaract Club of Deonar*

- Successfully organized on-ground Blood Donation Camps during the Covid-19 pandemic (First wave).
- Successfully organized donation drives for Maranatha Harvest Mission Orphanage and Bal Anand WorldChildren Welfare Trust where on a monthly basis essential items and groceries were donated.

Jun 2019 – Mar 2020
Mumbai, India

Teacher | *U&I Trust*

- Taught grade-level English to children at Our Lady's Home, Dadar a shelter for orphaned and destitute children.

Mar 2018 – Present
Mumbai and Coimbatore, India

Scribe for People with Disabilities | *Independent*

- Writer for individuals who are visually and/or physically impacted.

Mar 2015 – Present
Mumbai and Bengaluru, India

Lifetime O Negative Blood Donor | *Independent*

- 21 donations till date